



PAWAAC DRONES

HAWKAI PLUS UAV

LONG-ENDURANCE TACTICAL SURVEILLANCE PLATFORM

DESIGNED & MANUFACTURED IN INDIA

ADVANCED ISR PLATFORM

MILITARY & HOMELAND SECURITY

PRODUCT OVERVIEW

The **HAWKAI PLUS** is a compact tactical aerial platform for surveillance, reconnaissance, perimeter monitoring, and field-response missions.

Optimized for stable hover, autonomous execution, long-range observation, and reliable deployment in demanding environments.

MISSION CAPABILITIES

 Tactical Surveillance

 Reconnaissance

 Border Security

 Search & Rescue

 Disaster Response

 Critical Infrastructure



FEATURE HIGHLIGHTS



80+ Minutes

Endurance with Thermal Surveillance
Pod payload system



60+ Minutes

Endurance with Optical Pod Camera
payload system



15 km Range

Operational range with directional
patch antenna extension



AI Target Tracking

Artificial intelligence assisted target
locking and tracking



Thermal Imaging

640x512 IR sensor with multiple
pseudo-color modes



Laser Rangefinder

Precision distance measurement from
5m to 1000m



30x Hybrid Zoom

10x optical plus 3x digital zoom for
long-range detail



30-Second Swap

Quick-release payload interface for
rapid mission adaptation



GPS Autonomous

Waypoint navigation and
autonomous mission execution



Return-to-Home

Automated return and landing on
signal loss or command



45 Knots Wind

All-weather wind resistance for
demanding field conditions

DUAL INTERCHANGEABLE PAYLOAD ARCHITECTURE

Two payloads. One platform. Swap in approximately **30 seconds** without tools.

01 THERMAL SURVEILLANCE POD



- Thermal Camera — 640x512 IR sensor
- Day Camera — 5 MP visible imaging
- Laser Rangefinder — 5m to 1000m, +/-1m
- AI Target Tracking — Automated lock-on
- 80+ Min Flight Time — Extended endurance

MISSION PROFILE

- Night Operations | Border Surveillance
- Search & Rescue | Thermal Reconnaissance

02 OPTICAL POD CAMERA



- 4 MP CMOS Sensor — HDR imaging
- Starlight Vision — Low-light capability
- 30x Hybrid Zoom — 10x optical + digital
- 2K Recording — 2560x1440 @ 30fps
- 60+ Min Flight Time — Extended endurance

MISSION PROFILE

- Day Surveillance | Observation
- Inspection | Target Monitoring



-30 SEC
Payload Swap

THERMAL SURVEILLANCE POD

3-Sensor Stabilized Optical Pod | IP54 Rated

SPECIFICATIONS

Parameter	Specification
Stabilization	3-axis gimbal
Visible Camera	5 MP module
Recording	2880x1620 @ 30fps
Thermal Sensor	640x512 IR
Thermal Band	8-14 um, 30 Hz
Detection Range	Up to 1100 m
Laser Rangefinder	5m - 1000m
Ranging Accuracy	+/- 1 m
AI Functions	Target lock & track

D.R.I. PERFORMANCE

1100 m DETECTION	550 m RECOGNITION	275 m IDENTIFICATION
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Performance under favorable operating conditions. DRI = Detection, Recognition, Identification



80+ MIN

Flight Time with Thermal Payload

THERMAL DISPLAY MODES

White Hot | Black Hot | Multiple Pseudo-Color Modes

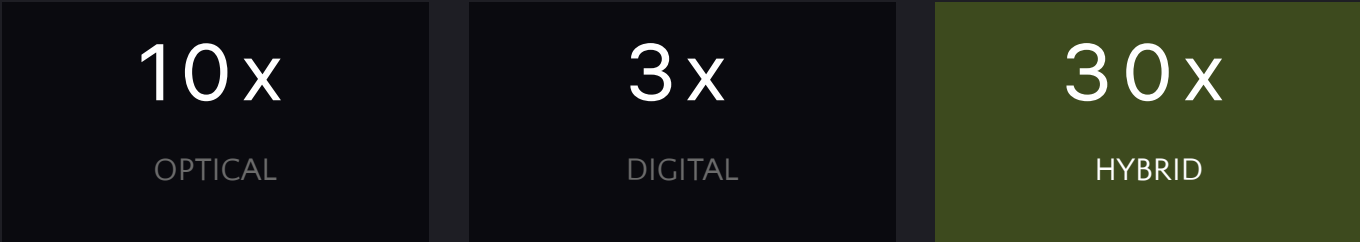
OPTICAL POD CAMERA

10x Optical Zoom | 3-Axis Stabilization | MAVLink / ArduPilot / PX4

SPECIFICATIONS

Parameter	Specification
Sensor	4 MP CMOS
HDR Imaging	Supported
Low-Light Vision	Starlight capability
Optical Zoom	10x
Hybrid Zoom	30x (10x Opt + 3x Dig)
Recording	2K 2560x1440 @ 30fps
Stabilization	3-axis gimbal
Compatibility	MAVLink / ArduPilot / PX4
Flight Time	60+ minutes

ZOOM CAPABILITY



30x Hybrid = 10x Optical + 3x Digital zoom combination



60+ MIN

Flight Time with Optical Payload

MISSION APPLICATIONS

Day Surveillance | Observation | Inspection
Reconnaissance | Target Monitoring

AIRFRAME SPECIFICATIONS

AIRCRAFT PARAMETERS

Parameter	Specification
UAV Type	Quadcopter UAV
All-Up Weight	2.4 kg (without payload)
Frame Size	510 x 510 mm
Propeller Size	15 inch
Endurance (Thermal)	80+ minutes
Endurance (Optical)	60+ minutes

PERFORMANCE PARAMETERS

Parameter	Specification
Operating Range	Up to 15 km
Range Extension	Directional patch antenna
Wind Resistance	Up to 45 knots
Standard Altitude	120 m AGL
Extended Altitude	300 - 500 m AGL (authorized)
Service Ceiling	Up to 4000 m AMSL

NAVIGATION & CONTROL



STABILIZE



LOITER



MANUAL



AUTONOMOUS



LAND

Ground Control Station based mission planning with waypoint navigation, position hold, autonomous route execution, telemetry monitoring, and controlled recovery functions.

CAMERA SPECIFICATIONS

THERMAL SURVEILLANCE POD		OPTICAL POD CAMERA	
Parameter	Thermal Payload	Optical Payload	
Sensor Type	5 MP + 640x512 IR	4 MP CMOS	
Recording	2880x1620 @ 30fps	2560x1440 @ 30fps	
Zoom	30x Digital	10x Optical / 30x Hybrid	
Stabilization	3-axis gimbal	3-axis gimbal	
Thermal Band	8-14 um, 30 Hz	N/A	
Detection Range	Up to 1100 m	Up to 1200m	
Laser Rangefinder	5m - 1000m, +/-1m	N/A	
Special Features	IP54	HDR, Starlight, MAVLink	
Flight Time	80+ minutes	60+ minutes	

SAFETY FEATURES & APPLICATIONS

BUILT-IN SAFETY SYSTEMS



Battery Failsafe
Auto RTH on low power



Geofence
Virtual boundary protection



Return to Home
Signal loss recovery



Emergency Landing
Controlled descent logic



Mission Recovery
Resume interrupted missions



Parachute
Optional recovery system

MISSION APPLICATIONS



Border Surveillance
Perimeter monitoring and territorial security operations



Counter Infiltration
Detect and track unauthorized border crossings



Perimeter Security
Critical facility and base perimeter protection



Search & Rescue
Thermal and visual search for missing personnel



Disaster Response
Rapid assessment and situational awareness



Industrial Security
Infrastructure and asset monitoring



Law Enforcement
Tactical support and evidence gathering



Strategic Recon
Long-range ISR and target acquisition



PAWAAC DRONES

Engineering India's Next Generation Tactical UAV Systems

CONTACT INFORMATION

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